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Leveraging the ConvNeXt-L Model for Tomato Leaf Disease Classification

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***Abstract* — Accurate identification and classification of disease symptoms on the tomato leaves, is a vital part of the development of phyllobiological health and complementary agronomical productivity. This study proposes a carefully crafted deep learning architecture which is designed to complement detection and categorization of the most common tomato leaf diseases. The methodology begins with a preprocessing step which is performed separately by a U-Net segmentation network, which isolates the salient regions of the leaves and eliminates non-essential artefacts. Subsequently, it extracts discriminative features using ResNet50 backbone which is famous for its ability to extract multi-scale hierarchical representations. The feature maps that are generated are eventually fed into a ConvNeXt-L classifier, which is responsible for differentiating a wide range of classes of diseases. The proposed framework is tested with a dataset of images of tomato leaf disease, taken from Mendeley, containing 18,000 images with a clear labeling. The integrated model has a high accuracy of 98.12%, which shows its good potential for reliable disease classification.In addition, the comparison with other architectures that are well-known like InceptionV3 and DenseNet121, the suggested approach gives superior performance in terms of accuracy and feature discrimination.**

Keywords — ConvNeXt-L, deep learning, tomato leaf diseases, classification, resnet50.

I. INTRODUCTION

Tomato (Solanum lycopersicum) is an important vegetable crop that is cultivated across the world due to its nutritional characteristics, commercial importance, and diversity as a food item and as a food processing component. It contains important vitamins, minerals and antioxidants that make human beings healthy. Tomato cultivation is a significant revenue generator among millions of farmers and a substantial part of the overall and national agricultural economy because of its brief development period, significant demand in the market, and capacities to adapt to varying climatic conditions. However, tomato farming is contingent on numerous factors which include climate, soil fertility, water, pests, and cultivation technology and, among others, the most critical challenge is the leaf diseases [1]. They pose a direct challenge to the photosynthetic processes, undermine the well-being of plants and reduce the harvest. The tomato plants are susceptible to fungal, bacterial and viral infection resulting into lesions, discoloration, and physiological stress. The diseases are very fast spreading over fields resulting in massive loss of yield, increased cost of production and supply chains are disrupted. These diseases should therefore be detected early and correctly so as to reduce the damage as well as to be able to manage the crop effectively [2]. Sample pictures of tomato leaf diseases are shown in Figure 1.

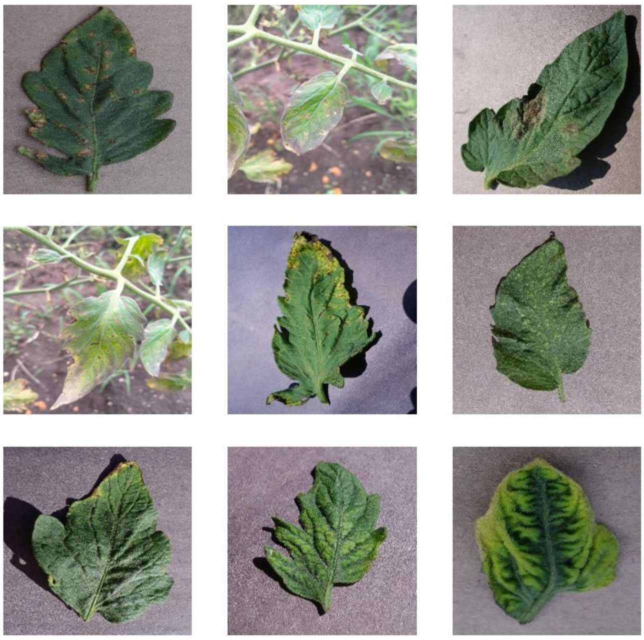

Fig. 1. Pictures showing the signs of tomato leaf disease.

Conventional disease diagnosis is primarily based on visual evaluation by the farmers or specialists. Even though the method has been in practice over decades, it is time-consuming, subjective, and cannot be applied to large-scale farming given that initial symptoms resemble those of various diseases. Lack of proper diagnosis and time lost in detecting the disease usually results in unchecked disease expansion and the overuse of pesticides, which lose their effectiveness on the environment and the incomes of the farmers. Deep learning has emerged as an effective identification tool of plant diseases to solve these challenges. Convolutional Neural Networks (CNNs) are able to automatically extract patterns of significance in images, which detect subtle shifts in color, texture, and shape that a person might overlook [3]. Deep learning systems are consistent, highly accurate, and able to process very large volumes of images at a fast pace, which is why it is appropriate in real-time agricultural implementation, which includes mobile diagnosis systems, tracking drones and automatic field surveillance. Nonetheless, images captured in the natural environment during agriculture are frequently noise-contaminated, shadowed, unpredictably lit, and disturbed by the background, which is why sophisticated architectures are essential to depend on the classification of diseases.

ConvNeXt, in this regard, has become an effective, contemporary convolutional architecture, which can be used to address real-world Agricultural image problems. ConvNeXt is a hybrid between traditional CNNs and transformer design concepts, which produces a potent but computationally efficient model [4]. This is because of its capacity to retrieve precise visual elements and therefore it is very efficient when it comes to the classification of the tomato leaf disease, where the disease spots, lesions and changes in texture can be minute and intricate. ConvNeXt uses better normalization, bigger receptive fields, and better convolutional blocks, allowing it to acquire rich and discriminative features even when the pictures have different illumination, angles, or direction of the leaves. As compared to most older CNN architectures, ConvNeXt is also resistant to noise and background clutter, a useful characteristic of field-collected agricultural data. It is also scalable, allowing its design to support the different models of its intended use making it suitable across a wide range of computing platforms, including cloud and mobile ones.

Due to these benefits, ConvNeXt can be successfully applied to the tomato leaf disease detection systems. It is an effective parameter of classification that can tell the difference between various disease types which have a thin line of distinction [5]. In combination with effective preprocessing and feature extraction methods, ConvNeXt can be a trusted instrument that promotes precision agriculture, which enables the identification of diseases in their early stages, minimizes the need for human observation, and enables farmers to make a wise decision related to the pests management and crop protection. With the continued transition of modern agriculture to digital and automated solutions, novel and powerful deep learning models such as ConvNeXt will constitute an important step towards sustainable, technology-based crop health monitoring.

The tomato leaf diseases pose a threat to agricultural productivity because they harm the health of the plants which translate to lower yield and poor crop quality. Conventional disease detection is dependent on physical examination of the areas by farmers or an agronomist. The method is tedious, subjective, and not always accurate particularly at the onset of the disease when the symptoms are usually mild or confluent with various diseases. Due to the delayed and inaccurate diagnosis, farmers can take action too late and use a lot of pesticides, and face greater economic losses. The need to identify and classify tomato leaf diseases has been urgently triggered by increasing demand of efficient crop management and by a more complicated disease pattern as a result of environmental variability. This paper will take care of this requirement by creating a deep-learning model capable of reliably identifying healthy and diseased tomato leaves. In this way, it is conducive to precision farming, and enhances disease management practices.

II. Related Work

Sakkarvarthi et al. in [6] proposed a convolutional neural network (CNN) to classify and identify tomato crop diseases. The CNN algorithm analyses more complicated leaf patterns, which allows it to draw the different types of diseases. Their method is able to deal with issues such as variation in the appearance of leaves and the varied manifestation of diseases thus it can be used effectively in agriculture on a real-time basis. The study can benefit precision agriculture by providing a reliable way to use automated means to monitor crop health that can increase yields and minimize losses.

Samal and Verma [7] came up with a tailored CNN that identifies and classifies diseases in tomato leaves. Their architecture is designed to be specific on fewer variables to accommodate overlapping symptoms and texture irregularities and is highly accurate with computational efficiency. This paper informs about the necessity of specialized deep-learning models that are aimed at exploring the aspects of plant disease detection shown in table.1

TABLE I. Analyses and Comparisons of the Relevant Works

Authors	Proposed Method	Key Features	Key Contributions
Sakkarvarthi et al. [6]	CNN-based framework for tomato crop disease detection	Processes complex patterns in leaf images; high precision and reliability	Offers an automated, real-time solution for monitoring tomato crop health, enhancing precision agriculture.
Samal and Verma [7]	Self-designed advanced CNN model	Handles overlapping symptoms; efficient and accurate disease detection	Tailored model addressing the unique complexities of agricultural datasets for tomato leaf disease detection.
Nawaz et al. [8]	DL framework with localization and classification	Identifies and categorizes diseased regions accurately; two-step approach	Enhances precision and reliability in disease detection through integrated localization techniques.
Shoaib et al. [9]	DL-based segmentation and classification	Isolates diseased regions for focused classification; handles subtle symptoms	Improves classification accuracy by integrating segmentation to optimize disease detection in plants.
Al-Shamasneh and Ibrahim [10]	Conformable polynomial image feature-based classification	Extracts unique mathematical features for high accuracy classification	Introduces advanced mathematical feature extraction for robust disease detection in tomato plants.
Agarwal et al. [11]	ToLeD: Simple CNN-based system for disease detection	Straight forward architecture; high accuracy and practicality	Balances performance, making it suitable for real-world agricultural deployment.

Attallah [12]	Compact CNN with transfer learning	Utilizes pre-trained networks; minimizes computational complexity	Develops resource-efficient models suitable for sustainable agriculture in resource-limited settings.

Nawaz et al. [8] came up with an end-to-end deep-learning model whereby the first model localizes disease regions on tomato leaves and the second model classifies the disease regions. They used two steps to localize and group diseased areas which resulted in a higher accuracy and reliability compared to split-step methods especially in a complex background or mixed symptoms. Their tool shows the usefulness of localization of diseases in enhancing the performance of the models in contemporary agriculture.

Shoaib et al. [9] planned a segmentation-and-classification structure of tomato leaf diseases. During the classification step, the model targets only the affected areas, through the preliminary segmentation that seals off the diseased areas. This enhances accuracy of detection, particularly of earlier symptoms. The research highlights the significance of segmentation in the fields of agriculture to enhance classification. A typing system called conformable polynomial image feature was introduced by Al -Shamasneh and Ibrahim [10] to classify tomato leaf features. High identification accuracy is obtained when the mathematical features extracted on images are used. They use the combination of advanced mathematical modeling and machine learning to tackle agricultural issues in an efficient and reliable manner.

Agarwal et al. [11] introduced the ToLeD, which is a CNN based system of detecting tomato leaf diseases. It has a simple, easily adoptable architecture, which is intended to work in farms. Although it is a simple algorithm, the ToLeD has high levels of identification accuracy, which implies its applicability in real-life. The article shows that the model is effective and performance-oriented in large-scale agricultural activities.

Attallah [12] proposed an approach that combines the small CNNs with transfer-learning and feature-selection in disease classification. Transfer learning also saves training time yet delivers good performance and feature selection identifies the most pertinent features at the lowest cost of computation. This paper highlights why tight and streamlined frameworks are required, particularly in the resource-strained environment to facilitate sustainable farming behaviors.

III. Proposed Method

To effectively classify tomato leaf diseases in this study, tomato leaf disease classification methodology is initiated with image preprocessing, defined as the U-Net encoder-decoder framework that is successfully utilized to reduce the fidelity of the image obtained with the view of identifying salient foliar loci and also reducing the effect of background perturbations. After the preprocessing, the ResNet50 convolutional neural network is used as a feature extraction backbone. Due to its 50 residual layers and the resulting skip connections, ResNet50 is well-equipped to learn complex, discriminative morpho-physiological features and avoid the notorious vanishing-gradient pathology, therefore, maintaining diagnostically salient patterns. The resulting feature embeddings are subsequently inferred into the ConvNeXt -L classifier, the state -of -the-art refinements of architectures and the increased representational capability of state-of-the-art classifiers provides a subtle differentiation of all tomato leaf diseases phenotypes. The unified framework in consequence provides a powerful, high-quality diagnostic modality, providing the stakeholder with an automated and scaled tool in the surveillance of crop health and providing the foundation of the more comprehensive data-driven precision agriculture paradigm.

A. Preprocessing

U-Net is an effective image segmentation model since it is able to capture fine spatial information and bigger picture [13] . Originally meant to be used in biomedical images it is today applied in numerous applications and is currently being used in agriculture to find and match diseased regions. To classify tomato leaf disease, U-Net is first used to segregate the leaf and its complex backgrounds (soil, shadows, or adjacent vegetation) such that only the leaf surface and areas of disease remain. U-Net isolates sick spots and eliminates noise, which supplies the classification model with sharp and concentrated data. It has multiple scale encoder-decoder architecture which learns rich features. The encoder is what reduces the image to abstract feature maps and the decoder reinstates these abstract feature maps into a clean segmentation without spatial resolution degradation. Skip connections between decoder and encoder layers do not lose fine details, including lesion borders, mottling, discoloration, and the shape of infections as it is. This renders U-Net suitable in the identification of fine or anomalous disease indicators on tomato leaves. The image segmentation generated by U-Net is more localized and finer. This increases the saliency of target areas and eliminates extraneous backdrop sceneries and differences in lighting, which may cause misinterpretation to the target downstream models. The narrow preprocessing minimizes the computational burden, improves the risk of misclassification and makes a system as a whole more resilient.

Precise segmentation is critical in agriculture since diseases present themselves in numerous forms such as specks, increasing spots of the blight, ring lesions, chlorosis, or even as fungus. The pixel-level discrimination of U-Net preserves such variations whilst preserving the natural structure of the leaf. The outcome is Early disease detection which is useful in preventing disease spread and minimizes loss of crops. The quality of better input also creates more dependable and interpretable disease-detection systems which are trusted by farmers and researchers. In general, the U-Net enhances the precision and efficiency of tomato leaf disease classification through the generation of clean, accurate, and informative segmented images. The improved preprocessing contributes to more successful disease management in precision agriculture by providing improved crop -health monitoring, allowing timely intervention and ultimately.

B. Feature extraction

One of the steps that are important in classifying tomato leaves is feature extraction since it enables the system to identify the variation between the diseased and healthy leaves. The model pulls out viable visual cues during extraction, including differences in color, differentiation in texture, lesion morphology, distortion of veins, and change of edges. These hints are the basis of making correct diagnosis. Together, the features that are extracted inform the classification model on the manner in which each disease presents itself at a tomato leaf. ResNet50 is an excellent method of deep learning classification models due to its ability to discover discriminative features in pre-processed images automatically [14]. The network consists of 50 layers connected by residual connections which conserve valuable information on forward and backward runs. Those connections ensure that important properties do not disappear as the model becomes more profound, addressing the vanishing -gradient issue that most deep neural networks have. Consequently, ResNet50 is able to maintain the learning state when it works with complex and high-dimensional agricultural images.

ResNet50 is the best at identifying minor variations in leaf texture, color gradation, and structure, which means that it is suitable in the identification of fungal, bacterial, and viral symptoms. The network can support a broad spectrum of the patterns of symptoms whether it manifests as tiny speckles, growing necrotic foci, or discolouration. The feature maps of ResNet50 provide multi-level representations, i. e. simple edge and shape features in the lower layers, and disease-specific features in the higher layers. The extracted high-quality features pass to the classification stage. Patterns that are studied by the classifier include surface roughness, size and shape of lesions, degree of chlorosis, and distortions of boundaries that are used to separate disease categories. Accuracy directly depends on the quality of obtained features. Discriminative features that are well captured enable the system to exactly predict even in case of similarity in diseases that appear similar.

A powerful extractor such as ResNet50 provides some form of uniformity among the images captured at various times of lighting, angles or leaf directions. This consistency increases the generalization of the model that allows it to work on both field-collected and lab-controlled images. Extracted features also make the interpretation easier; activation maps can display which leaf regions the model considers the most relevant in its decision. Practically, these discriminative cues should be regarded as properly extracted and utilized. Once the system is able to identify disease-specific visual cues with great accuracy, farmers will be able to react in time, administer specific treatments, and avoid the further transmission. This enhances the management of crops, reduces economic losses and promotes the objective of precision agriculture. Finally, successful feature extraction is the foundation of any high-performance model of disease-classification and is essential to the creation of automated tools that can help maintain sustainable crop protection.

C. Classification

ConvNeXt-L is the last classifier that will be used in this paper, and it will be used to differentiate the tomato leaf disease types and deep features of processed images [15]. The image is first improved using U-Net and then discriminative features are extracted using ResNet-50. These highly refined representations are inputted into ConvNeXt -L that discovers fine-grained and high-level patterns of diseases. Large-kernel depthwise convolutions, normalization layers, GELU activation, and residual connections are included to the architecture to make sure that the disease-specific cues, lesions, chlorosis, fungal spots, and texture distortions, among others, are well captured. The features are transformed by ConvNeXt -L and then a fully connected layer followed by a Softmax function classifies the images [15]. The output scores are transformed into probabilities of the classes using the following formula:

$$P(y = k \mid x) = \frac{e^{z_k}}{\sum_{j=1}^{K} e^{z_j}} \tag{1}$$

In this case, z_k will be the score of the model on the output of class k. We use cross-entropy loss to be optimal in predicting tomato leaf disease during training, whereby mistakes are punished and the model is reinforced to make distinguish diseases:

$$L = -\sum_{k=1}^{K} y_k \log\big(P(y = k \mid x)\big) \tag{2}$$

In this case, the ground-truth class label is referred to as y_k. The model after training is evaluated by measuring the accuracy of the model that differentiates the diseases-Early Blight, Late Blight and Septoria leaf spot as well as healthy leaves using the normal performance measures.

Besides the core classification ability, ConvNeXt -L has architectural enhancements that can be used to benefit the suitability of the system in actual agricultural image analysis. The convolutional kernels are large and the model is able to pick the long-range spatial structure, this aids in detecting the patterns of disease that are propagating along veins or along leaf margins. Layer Normalization is used to stabilise training so as to minimise the sensitivity to lighting differences and background noise typical of images in the field. ConvNeXt -L may intrinsically learn more detailed and richer feature representations with the assistance of the channel -expansion mechanism that allows the model to distinguish between diseases with similar colour tones or texture patterns shown in fig.2.

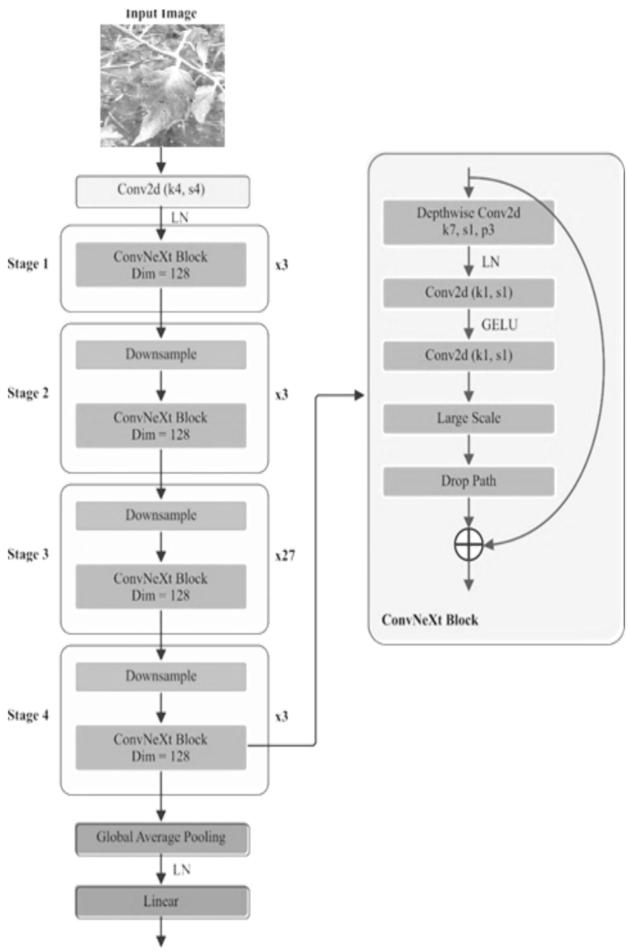

Fig. 2. ConvNeXt-L architecture.

ConvNeXt-L has good gradient flow and well-constructed residual connections, so key attributes of a disease, like small circular lesions, marginal burns or an irregular spot of fungi, are not lost to deeper processing. Its hierarchical learning strategy allows the model to develop over time through simple edge and texture detection to more abstract representations of disease to increase its discriminatory ability. The network is also capable of supporting changes in image angle, size, orientation, brightness and disease severity hence suitable to real world tomato leaf datasets.

ConvNeXt-L has sensible advantages in the field. Although it is computationally efficient, the model is still deep enough and can be embedded into mobile apps, drone systems, and farm-based devices to monitor in precision agriculture. This renders the diagnosis of diseases available at remote areas where consultation by the expert is minimal. ConvNeXt -L enables farmers to identify diseases at the initial stage, implement specific treatment and reduce the needless use of pesticides by offering reliable and similar classification results. This eventually serves to assist in smarter crop management, economic loss reduction and sustainable agricultural practices.

IV. Results

A. Data Set

The dataset of the Tomato Leaf Disease [16] consists of 18,000 tomato leaf images. In every picture, a healthy leaf or a diseased leaf with a particular disease is presented. The data is separated into ten different categories which involve healthy samples and most common tomato leaf disease like bacterial spot, blight, leaf mold and other fungal and viral infections. Each category contains equal the number of images, and it ensures that the distribution of classes is even and less bias in training. The photos were captured under varying environmental settings. They differ in the light, the background, the positioning of leaves as well as the severity of the disease. This diversity assists machine-learning and deep-learning models to make inferences on the real-world scenario. The data is rich in different textures, colors, and patterns of symptoms, which allow creating high-quality disease-classification models. The Tomato Leaf Disease Dataset is therefore a great standard in assessing the performance of models and in coming up with machine tools that can help in precise identification of the disease and crop development.

B. Evaluation

Tomato Leaf Disease Dataset can be successfully trained and tested on an 80:20 split. Eighty percent of the pictures will be used as an input during the training process and 20 percent will be used to test the performance of the model. The training set presents the deep learning model with a great extent of tomato leaf disease patterns. It assists the model to learn and know the distinguishing features of each of the classes in terms of visual features. These involve variations in the shapes of the lesions, variation in color, and distortion of texture and dispersion of the infection on the leaf surface. The model builds a powerful combination of disease-specific features by training on a vast number of labeled images and this is critical in proper classification.

The model is tested on testing set after the training process. This set includes the images which the model never saw. It guarantees a fair evaluation of the model in terms of its capacity to be applied to new data and avoids memorizing or overfitting. The analysis is done based on conventional classification measures like precision, recall, and accuracy, and F1-score. These measures can give a complete picture of how the model is effective in identifying diseased leaves, and differentiating them with healthy leaves, and whether it is consistent across varied categories. Precision is an indicator of the precision of positive prediction. Recall assesses the detection capability of the model on the true diseased cases. Accuracy provides a general measure of correctness. F1-score is a balance between precision and recall and is a single figure that is the sum of these two values.

The use of these metrics would guarantee an extensive verification of the practical performance of the model. It assists in its preparation to actual field farming. A model that is predictable in all the measures indicates strength and trustworthiness. This performance renders it very appropriate in assisting farmers to detect diseases early and make decisions. On the whole, all these evaluation practices provide a strong belief in the application of automated deep learning-based systems as effective means of modern precision agriculture.

$$Accuracy\ (Accy) = \frac{TrP + TrN}{TrP + FlP + FlN + TrN} \qquad (3)$$

$$Precision\ (Pr) = \frac{TrP}{TrP + FlP} \qquad (4)$$

$$Recall\ (Rl) = \frac{TrP}{TrP + FlN} \qquad (5)$$

$$F1\,Score\ = \frac{2 * (Pr * Rl)}{(Pr + Rl)} \qquad (6)$$

TABLE II. ANALYSES AND COMPARISONS WITH PREVIOUS WORKS

Model	Accuracy	Precision	Recall	F1 Score
Inception V3	90.26	83.23	91.44	84.23
VGG 16	92.46	88.86	90.86	80.45
DenseNet 121	95.64	90.11	90.18	80.87
Proposed Model	**98.12**	**94.02**	**91.96**	**92.21**

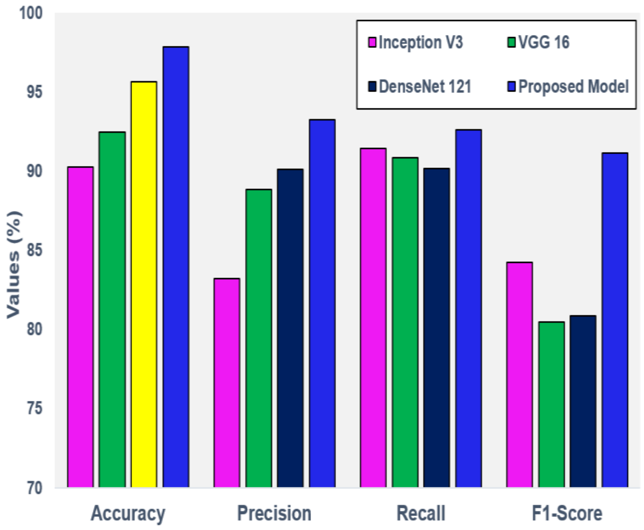

Fig. 3. Comparisons of proposed model performance.

Figure 3 and Table 2 present the bar diagram indicating the performance of four classification models. The accuracy of the proposed model is the greatest at 98.12. This finding shows that it is better at detecting and categorizing tomato leaf diseases. It is also highly precise (94.02%), recalls (91.96%), and F1-score (92.21%), which means that it performs well in both directions and has a small false prediction rate. A combination of these metrics proves that the proposed strategy is reliable and consistent in all the evaluation parameters. The second best model in the other models is DenseNet121 as it has an accuracy of 95.64. It is also very good but not as accurate as the proposed system. VGG16 and InceptionV3 obtain moderate accuracy of 92.46 per cent and 90.26 per cent respectively. The fact that they scored lower indicates that they are not as much able to learn the complex features of disease-specific in tomato leaf images. On the whole, the comparison indicates that proposed model is the strongest and effective among all the tested models. It is an efficient tool that should be relied upon to reliably detect tomato leaf diseases because of its high classification performance, high precision, high recall, and high F1-scores. This device can develop accuracy farming and enhance crop health control.

V. CONCLUSION

The proposed deep learning model has the ability to classify tomato leaf diseases accurately and based on the visual traits of these diseases. The system achieves an accuracy of 98.12% by using the ResNet50 as the feature extractor and ConvNeXt-L as the classifier, showing that the system has great potential in the recognition of different disease patterns. Such a degree of performance underlines the promise of the model being a useful instrument in precision agriculture, and that it can provide a reliable diagnosis of diseases that can help to produce healthier crops and yield more results. The results indicate that this kind of automated technique can be of great help to the farmers because the reliance on manual inspection is minimized, and prompt action can be taken. The further work will be aimed at implementing the suggested tomato leaf disease classification model in edge computing systems. This will facilitate real-time and on-field detection of the disease. The connection of the system to edge devices, including smartphones, drones, and IoT-based agricultural sensors will minimize reliance on the cloud infrastructure. It will also reduce latency which will enable farmers to get immediate diagnostic feedback even where there is poor internet connectivity. In order to implement it on resource-limited devices, we are going to consider model optimization techniques. These are pruning, quantization and lightweight inference approaches. The extension will increase the practicality, responsiveness, and scalability of the system, which will render the technology more appropriate to the real-life precision agriculture use and uninterrupted crop health observation.

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